LING 406: PSYCHOLINGUISTICS

“Normal” Speech Errors: Part I
Session 7

Dr. Jordan Lopez

Before we amuse ourselves with a catalogue of different kinds of speech errors, let’s consider:

Why do people make speech errors?

Why should they be of particular interest to psycholinguists?

Fromkin’s article is titled “Speech errors as linguistic evidence.” Why is that kind of a gutsy thing for a linguist to call a paper?

Key concepts for understanding how speech errors happen in terms of “spreading of activation” models:

“Noise” – not just literal, acoustic noise (although this can cause problems, too), but noise related to the *activation* in different parts of the system.

Frequency – remember from our discussion of “how the brain works” that more common words are easier to “activate” than less common ones. This seems to impact many different levels of speech production.

Other sources of activation – we have to assume that what you’ve just said and what you’re planning to say are active to some extent while you’re talking. There may also be activation from things you’re thinking about but trying not to say…

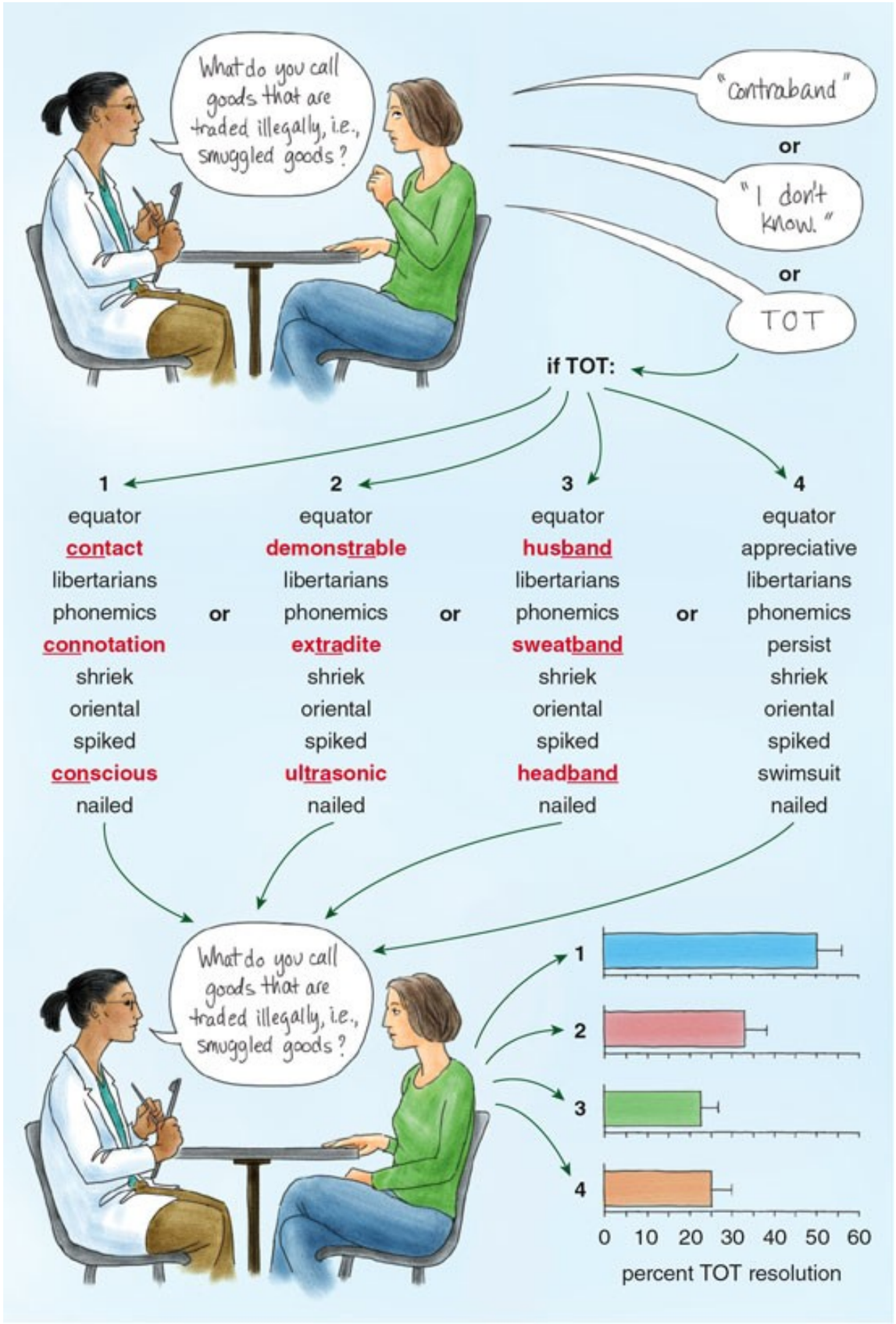
Similarity – activation spreads more easily between related items. Similarity can be semantic, syntactic, phonological, etc…

Tip of the tongue (TOT) states:

We might describe a TOT state as partial activation of the semantic and phonological forms of a word, that falls short of the threshold for producing the word…

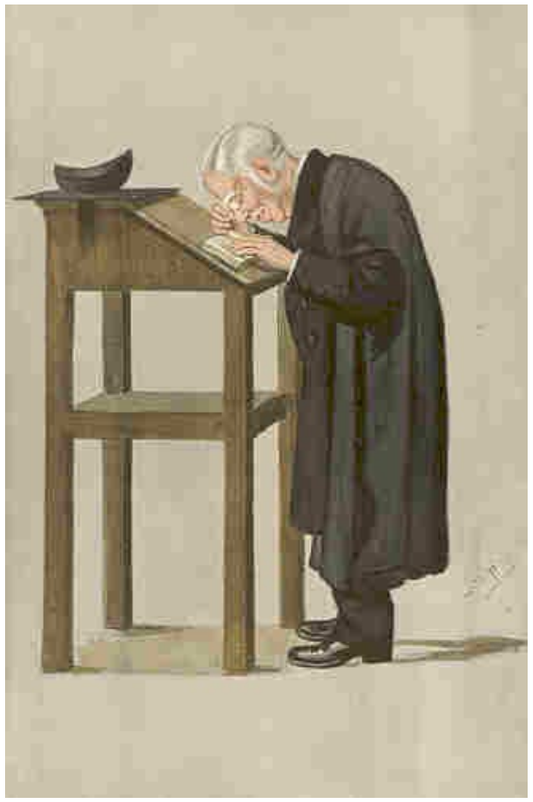
What kinds of words are susceptible to TOT?

Why might priming help?



Graphic from this American Scientist review by Lise Abrams: http://www.americanscientist.org/issues/id.3626,y.2008,no.3,content.true,page.3,css.print/issue.aspx

Slips of the tongue



Dr. Spooner who supposedly once said:

“Mardon me padam, this pie is occupewed. Can I sew you to another sheet?”

anticipation

(1) (a) John dropped his cup of coffee → … cuff… coffee
(b) also share → alsho share [ɔlšo šer]
(c) such observation → sub – such …
(d) delayed auditory feedback → … audif – auditory …
(e) week long race → reek long race
(f) M-U values [ɛm juw væljuwz] → [ɛm vjuw] values
(g) the third and surviving brother → the sird and – the bird – the third …

perseveration

(2) (a) I'm not allowing any proliferation of nodes → … proliperation
(b) John gave the boy → … gave the goy
(c) Spanish speaking people → … speaping people
(d) irreplaceable → irrepraceable
(e) Chomsky and Halle → Chomsky and Challe

metathesis

(3) (a) keep a tape → teep a cape
(b) the zipper is narrow → the nipper is zarrow
(c) should serve → [sʊd šərv]
(d) for far more → for mar fore
(e) with this ring I do wed → … wing … red
(f) I'm going to die young but I'll die less young → … yes lung
(g) in the past few weeks → … fast pew [pjuw] weeks.

All examples from Fromkin, in the optional reading. See also: https://en.wikipedia.org/wiki/William_Archibald_Spooner

Slips of the tongue

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What aspects of our "spreading activation" framework are most obviously at work here?

Noise?
Frequency?
Other sources of activation?
Similarity?

“Slips of the ear:” Mondegreens

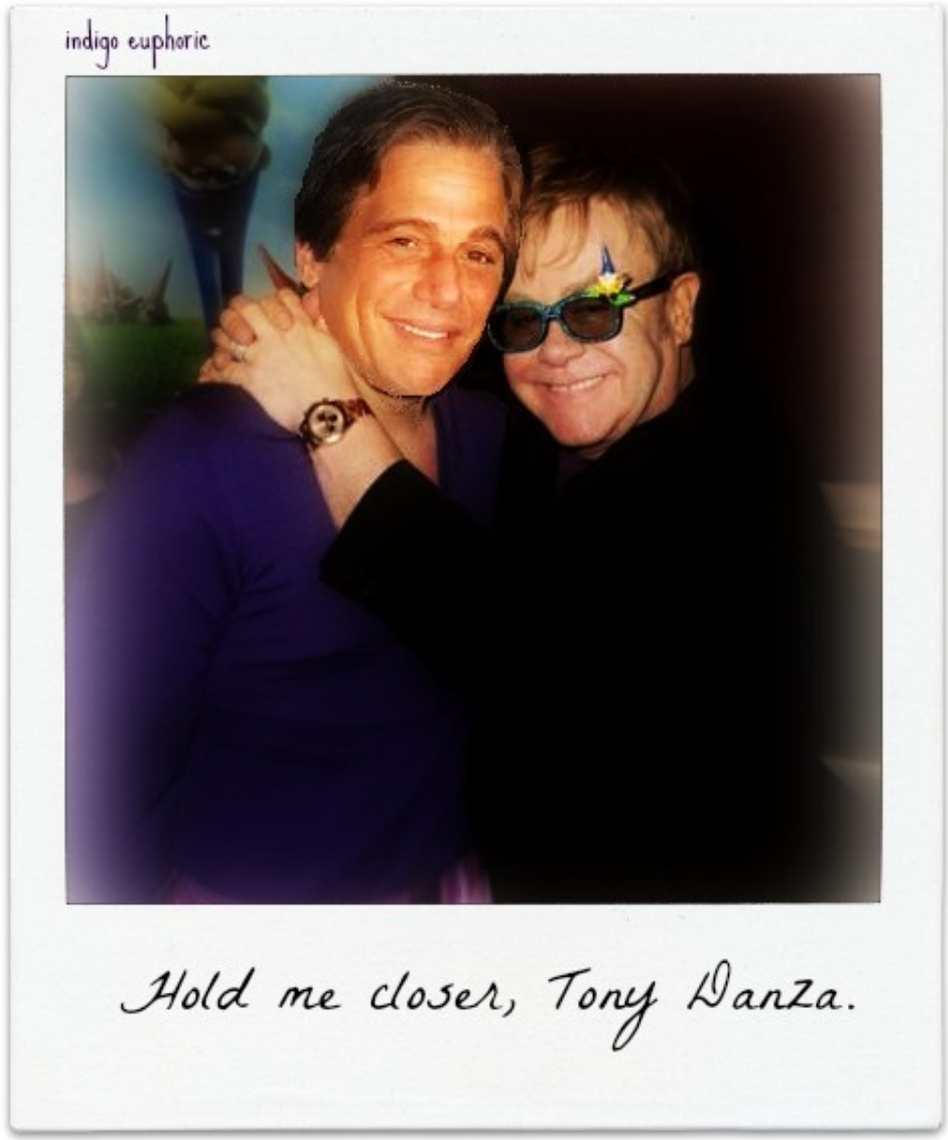
The term “mondegreen” itself comes from a mishearing of “laid him on the green” as “Lady Mondegreen.”

Remember when we discussed the ambiguities of speech, and word boundaries?

Consider how you would disambiguate:

“The good candy came anyways” from
“The good can decay many ways.”

Context can be very important, especially in songs, where prosody is less helpful.



For the record, it’s “Tiny Dancer,” like the title of the song…

Andrew Nevins is using crowdsourced data from kissthisguy.com to do analyses of this: http://www.ucl.ac.uk/llsd/projects/Slips_of_the_ear

More slips of the ear: Malapropisms

"Observation from one of my preschool students after removing her winter hat: 'My hair's ecstatic!'"

Barbara Donlon

"An old friend, recalling the good times we had as youths, fondly called that time our 'lettuce' days."

Daniel Schaffer

"A young friend of mine describing a scene from a horror movie she had just seen: 'A giant knife flew threw the air at the guy and decaffeinated him.'"

saintroch

"A friend described her father as having a congenial heart defect (congenital, of course, and decidedly not friendly)."

Constance Ruth Price

"My aunt always said, 'I can't have anymore children because I've had my Utopian tubes tied.'"

J. Toscano

"As I left work one Friday evening, I told the security guard on duty, 'Have a good weekend!' 'Clockwise!,' he responded cheerily."

jverra

These aren't so much *speech errors,* as *perceptual errors,* because the speaker *intends* to say just what they've said.

The speaker seems to have heard a word like "congenital," and decided that what must have been intended was some other, more familiar word (e.g., "congenial").

Again, we see top-down processing at work…

From: http://schott.blogs.nytimes.com/2010/02/01/to-all-intensive-purposes/

Using speech errors as data: (more from Fromkin...)

3.2. CLUSTERS AS SEQUENCES OF DISCRETE PHONES OR SEGMENTS. The above examples show errors of transposition, substitution, omission, and deletion of individual segments, which may be either vowels or consonants. The error may be either of anticipation (i.e., the interfering segment follows the error), of perservation (i.e., the interfering segment precedes the error), or of transposition (i.e., the order of sound segments is changed). Further justification for assuming that individual segments are units in speech performance is suggested by the fact that, in many errors where the intended utterance included consonant clusters, only one segment of the cluster is involved:

(6) (a) fish grotto → frish gotto
(b) fresh clear water → flesh queer water
(c) split pea soup → plit spea soup
(d) brake fluid → blake fruid
(e) no strings attached → no strings attrached
(f) at the Broadway stores the prices are → ... spores the prices are
(g) in a split second → ... slit second
(h) that's a sticky point → ... spicky point
(i) a car with a stick shift → ... [št'k sɩft][4]

At the time this paper was written, there were arguments about whether the phoneme (or phone) was a unit in speech planning.

It might have turned out that whole clusters *always* move together, which would have yielded, e.g., *gish frotto (6a),* or *clesh freer water (6b)*...

Instead, Fromkin presents evidence that clusters get broken up all the time in production.

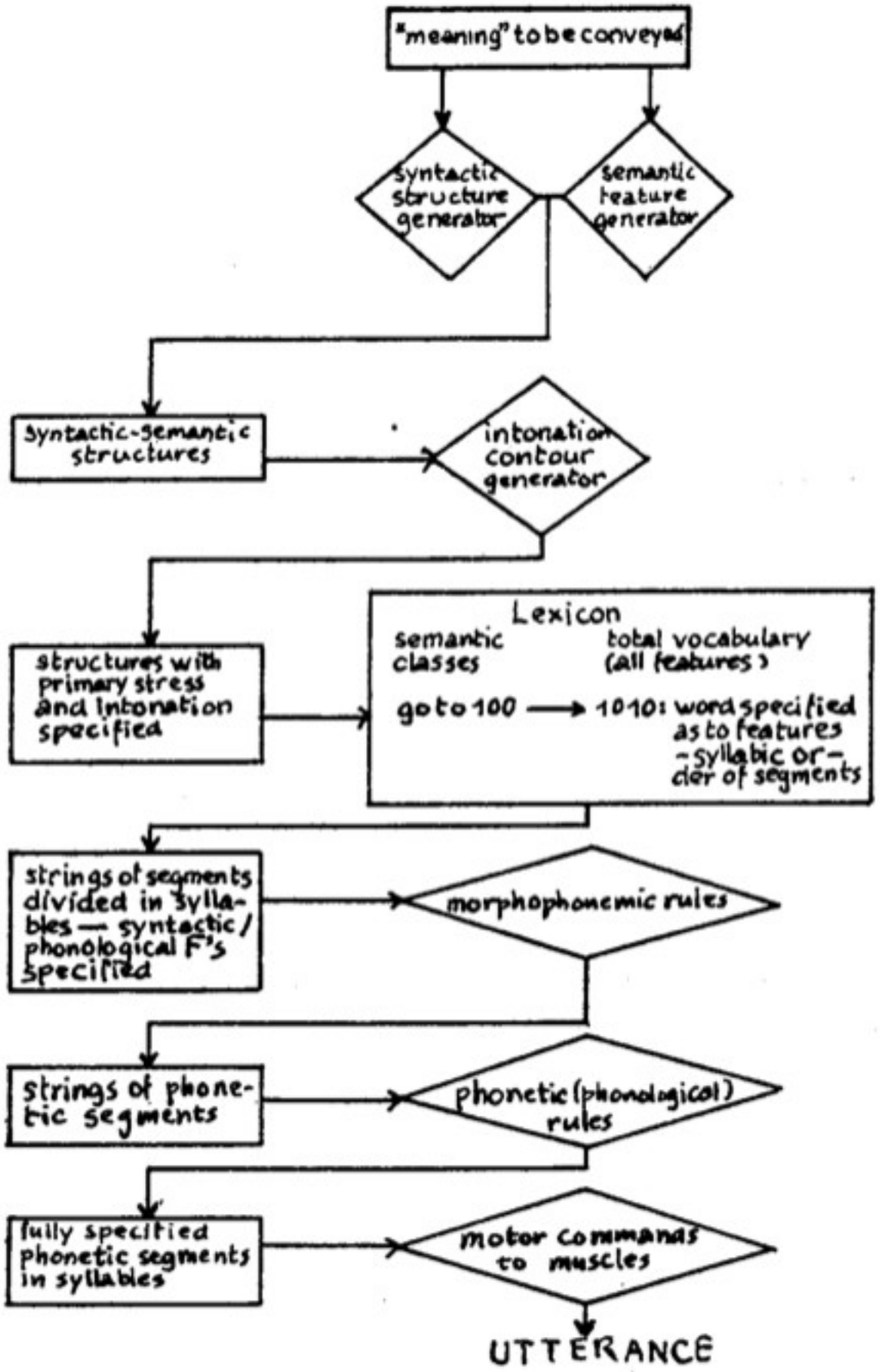


Fig. 1. Utterance generator.

An early version of the model Menn describes (from Fromkin, 1971)

Components of a “basic model” of speech production:

Levels:
Message
Functional
Positional
Phonological
Encoding
Speech Gesture

Following Menn, we’ll discuss only the first two today.

(Although we’ve already seen some phonological/encoding errors.)

The Message Level

"I won a bunch of money on the game Sunday."

How would you express this:

1. In a conversation with someone who was rooting for the losing team.

2. In a conversation with someone who is not interested in football.

3. In a tweet.

The Message Level is actually sort of mysterious, and hard to investigate experimentally.



Some speech production studies try to control the message level with an *elicitation paradigm*, like the picture book “Frog, where are you?”

But it’s still hard to define what the intended “message” is in a scene like this one, or how construction of that message may depend on age, culture, audience, etc.

The Functional Level -- Lemmas

We can’t just say that we select the “words” that we’re going to produce at the functional level…

Why not?

Consider this error:

Where’d I put it when I brought it in → Where’d I bring it when I put it in.

So, the verbs switched places, but the actual words that were produced reflect a change in tense that must have been applied later.

This couldn’t have happened if the *word* “brought” was selected. Rather, we say the lemma BRING was selected, and was expressed as present tense…



No, not Lemmings.

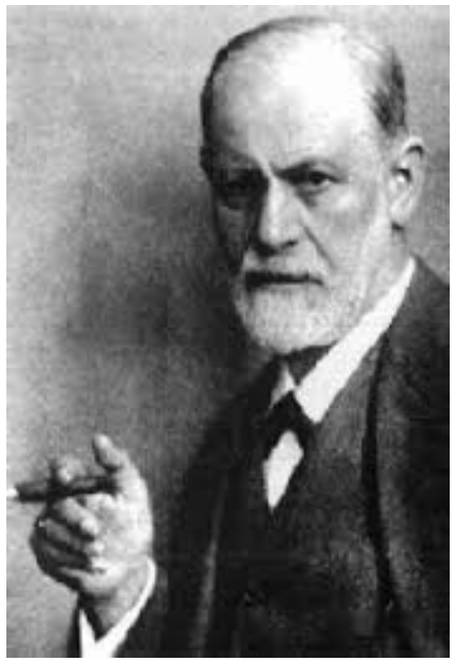
(The term “lemma” is borrowed from mathematics, where it refers to a **proposition** in an **argument**.)

And they *do not* just follow each other blindly off cliffs, BTW.

Errors in lemma selection:

Why do people so often say the opposite of what they mean?

(30) (a) I really like to – hate to get up in the morning
(b) It's at the bottom – I mean – top of the stack of books
(c) This room is too damn hot – cold
(d) the oral – written part of the exam



Let's leave this guy out of it.

The Functional Level

Why do people so often say the opposite of what they mean?

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Antonyms are actually quite *similar* at a number of levels:

- semantically related
- strongly associated

… it's like a perfect storm in the spreading activation framework…

further:

- can take on the same semantic roles
- same word class (can take on same syntactic roles)

…so the rest of the speech production system can handle them…

So it's easy for activation to spread from a lemma to its antonym.

Blend
s

[3] Lewis Carroll, in his preface to *The hunting of the snark* (1876), discusses his 'portmanteau' words: "... let me take this opportunity of answering a question that has often been asked me, how to pronounce 'slithy toves'. The 'i' in 'slithy' is long, as in 'writhe'; and 'toves' is pronounced so as to rhyme with 'groves'. Again, the first 'o' in 'borogoves' is pronounced like the 'o' in 'borrow'. I have heard people try to give it the sound of the 'o' in 'worry'. Such is Human Perversity ... Humpty-Dumpty's theory, of two meanings packed into one word like a portmanteau, seems to me the right explanation for all. For instance, take the two words 'fuming' and 'furious'. Make up your mind that you will say both words but leave it unsettled which you will say first. Now open your mouth and speak. If your thoughts incline ever so little towards 'fuming' you will say 'fuming-furious'; if they turn, by even a hair's breath, towards 'furious', you will say 'furious-fuming'; but if you have that rarest of gifts, a prefectly balanced mind, you will say 'frumious'."

I have quoted extensively from Lewis Carroll, not only because it is always a delight to read or reread any of his comments, but because in this passage he states that his 'portmanteaus' or 'blends' are possible in natural speech and proposes a hypothesis as to how they occur. As we shall see below, however, these 'complete' blends are seldom found in just this way in 'normal' speech. That is, a blend of [fjurijəs] and [fjumiŋ] is more apt to occur as [fjumijəs] or [fjuriŋ], particularly because the first syllables are identical.

Blend
s

(29) (a) My data consists [mownlij] – [mejstlij] … (mainly/mostly)
(b) I swindged [swɪndžd] … (switch/changed)
(c) It's a lot of [ba] – [brʌðl̩] (bother/trouble)
(d) She's a real [swɪp] chick (swinging/hip)
(e) it's a [spajrətɪv] (spirant/fricative)
(f) a tennis [æθlər] (player/athlete)

Such errors seem to support Carroll's assumptions. A speaker has in mind some meaning which he wishes to convey. In selecting words, it appears that he is matching semantic features. Where there are a number of alternative possibilities, rather than making an immediate selection, he brings them both into a buffer storage compartment, with their phonological specifications. Either a selection occurs at this point, or the words are blended, resulting in the above kind of errors.

(Lewis Carroll would have been an awesome linguist.)

Consider: "Where there are a number of alternative possibilities…"

When are there not a number of alternative possibilities in speech production?!

The Functional Level

Functional (Semantic) roles.

Remember:

{The plums} {in the icebox} were eaten {by me}
{Undergoer} + {Location} + Verb + {Agent}

Alternatively:

I + have eaten {the plums} {in the icebox}.
Agent + Verb + Undergoer + Location

So, we have a simple *event structure*, and we can start slotting lemmas into semantic roles. We know that the Verb will be TO EAT, the Agent of the verb will be ME, and that PLUM will be the Undergoer, etc.

Note there are syntactic consequences for arranging these roles in different orders, with consequences for how we turn the lemma for ME into a word.

That will happen at the *Positional Level*, in our next session…

At the functional level:

We have decided on a message to convey.

We have selected appropriate lemmas.

We have associated the lemmas with semantic roles.

…this is quite a bit, and yet we’re nowhere near putting a clause together, much less saying it out loud.

Required before Session 8:

Read Menn, Chapter 4 (pp. 133-151)

Optional follow-up:

Fromkin's introduction to "Speech Errors as Linguistic Evidence" is a classic: http://129.237.66.221/P800/fromkin.pdf
Most of the screen-grabbed text on these slides is from this paper.

On mondegreens:

Spotify's top ten: http://edition.cnn.com/2013/07/11/tech/web/spotify-misquoted-lyrics/

And this website devoted entirely to collecting them: http://www.kissthisguy.com/
(Note that this has a lot of user-generated content, and is bound to include some things that are offensive in a variety of ways.)

KEEP THE SPEECH ERRORS COMING!

NEXT SESSION WE'LL TRY TO CLASSIFY THEM...

Errors, humor and style

Very often speech errors go undetected. We will eventually talk about a lot of top-down effects in speech perception that explain this.

(Indeed, the process is similar to what we assume must have happened for many malapropisms or mondegreens)

Some “errors” are calculated for stylistic reasons.

Why might a deliberate error be more memorable than its correct counterpart?

Suddenly the old Will Rogers line — I’m not a member of any organized political party,I’m a Democrat — has lost its sting; the upper hand is on the other foot. And that’s going to color narratives and shape campaigns for a long time.



Paul Krugman: http://krugman.blogs.nytimes.com/2013/10/05/shorting-out-the-wiring/?_r=0

...quoting “Hot Shots: Part Deux.”